

Lab 16: Population & Systems Ecology (Module 15)

BIOL-1

Name: _____ Date: _____

Learning Objectives

By the end of this lab, you should be able to:

1. Interpret **exponential vs logistic** growth from verbal and tabular data, including **where per-capita growth is highest** along an S-curve.
2. Classify **density-dependent vs density-independent** regulation and connect each to **management levers** (what you can and cannot move).
3. Assign **trophic roles** and trace **energy** and **nutrients** through an ecosystem, including decomposers.
4. Apply the **~10% energy transfer** rule across multiple steps and explain **food-chain length** and **human food choices** in those terms.

Overview

You are the **acting manager** of the fictional **Green Mesa National Wildlife Refuge**. Elk were reintroduced in 2010; your field team sends **population notes** and **energy budgets** (invented but numerically consistent). You will **forecast** growth, **diagnose** regulation, compare **trophic “pyramids,”** and **audit** energy through a food web—work a reserve team might use to set harvest, fire, and habitat priorities.

Materials Needed

- Pencil or pen
- This worksheet
- Calculator (optional)

Part 1: Elk Come Home — From Boom to Ceiling

Elk counts (made-up):

Year Adult elk Notes

2012	40	Few predators; abundant browse
2014	90	Still rising fast
2016	155	Growth slowing
2018	198	Near stable
2020	205	Fluctuates ± 5

Δ per interval (adults/yr, rounded): 2012→2014: +25/yr; 2014→2016: +32.5/yr; 2016→2018: +21.5/yr; 2018→2020: +3.5/yr (use if helpful).

1. Shape of growth: Which curve best fits 2012–2018—**exponential (J)** or **logistic (S)**? **Circle one** and cite **two** observations from the table (or the Δ line) that support your choice.

2. Where growth is “fastest” in an S-curve: On a logistic curve, **per-capita** growth rate is typically **highest** when the population is **well below** K and **lower** as N approaches K . Does the 2012–2018 pattern match that story—**yes** / **no**, and **one** sentence of support using years or numbers.

3. Carrying capacity: From the table, estimate **K** (a range is fine, e.g., “about ____ elk”). What **resource** is most likely limiting elk if the story continued? Name **one** management action that could **raise** K for elk without changing the climate (short phrase).

- Estimated K :
- Limiting resource (hypothesis):
- One way to **increase** K (habitat or food side):

4. Prediction: If managers **remove half the elk** in 2021 and browse recovers, what do you predict about **population growth rate** the next few years (not just total number)—**increase** / **decrease** / **stay same**? Tie your answer to N relative to K in **one** sentence.

Part 2: Regulators, Management, and the Trophic Ladder

Three events; same big ideas: who limits the herd and where energy goes.

Event A: A **parasitic fly** spreads; infection probability rises as elk crowd into the same valleys.

Event B: A **wildfire** burns 40% of the refuge regardless of elk density.

Event C (new plant invader): A non-native shrub **outcompetes** bunchgrass; total **forage** for elk drops until crews remove the shrub, even at low elk numbers.

5. Classify each as density-dependent, density-independent, or mainly a resource/food limit (choose the **best** label; some cases blur—defend in one short phrase if you need to).

Event	Type (best label)	One reason
Event A		
Event B		
Event C		

6. Management puzzle: For **which** event is **herd size** a useful dial for reducing harm (alongside other work)? For **which** is the main tool **not** “shoot fewer or more elk” but **fire / vegetation / invasives**?

- Herd size matters most for:

- Mostly habitat/weather/invasion tools:

7. Why are apex predators numerically rare **even when prey are common**? (energy + numbers: **2–3** sentences)

Bridge: **Trophic levels** connect **who limits whom** (Part 2) to **how much energy** reaches the top of the food web (Part 4).

Part 3: Web Assembly — Who Eats Whom?

Refuge food web (simplified):

- **Sunlight** → bunchgrass & shrubs (**producers**)
- **Grasshoppers** eat plants
- **Ground-feeding birds** eat grasshoppers
- **Hawks** eat birds
- **Coyotes** eat elk calves (sometimes) and scavenge carrion
- **Fungi & bacteria** decompose dead material

8. Build one linear chain with four trophic levels (producer → ... → top carnivore). Use arrows **from food to eater** (e.g. grass → grasshopper → ground-feeding bird → hawk).

9. Label roles for the chain you wrote (producer, primary consumer, secondary, tertiary). If **coyotes** are in a **different** chain than your hawk chain, name **coyote's** most likely **trophic** role here (**secondary** or **tertiary** consumer vs scavenger) in one line.

10. Pyramids (concept check): A **pyramid of numbers** can look “wrong” (many small herbivores on one big tree) while a **pyramid of energy** is usually **upright** for the same system. In **2–3** sentences, explain **why** energy pyramids are the usual standard for “how much life can a habitat support at each level.”

11. Nutrients vs energy: In **2–3** sentences, explain why **decomposers** matter to **producers** even though energy does **not** cycle from dead matter back to sunlight.

Part 4: Energy Audit — The ~10% Rule

Use ~10% transfer between trophic levels (textbook-style rounding). Each step, about one-tenth of the energy in the previous level is available at the next—round to sensible digits and show brief work if the instructor requests it.

Given: Plants fix **1,000,000 kcal** of chemical energy in tissues eaten by herbivores in a season.

12. Step the energy down the chain:

Step	Available energy (kcal)
To primary consumers (herbivores)	
To secondary consumers	
To tertiary consumers	

13. From the same plant base, roughly how many kcal of **tertiary** consumer could be supported for each **1 kcal** of **herbivore** if you **stopped** the chain at herbivore? (A **ratio** in words or scientific notation is enough—e.g. “about one ____.”)

14. Sanity check: If humans tried to feed the world only on **apex tuna**, would we need **more** or **less** ocean primary production than if we ate **plants** (or **herbivorous** fish) directly? Use **2–3** sentences and the table logic, not a moral judgment about diet.

15. One-sentence contrast:

- **Energy through ecosystems:**

- **Nutrients through ecosystems:**